RSVMamba for Tree Species Classification Using UAV RGB Remote Sensing Images

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***Abstract*— Effective forest tree species (TS) classification is critical for various application domains such as forest management, biodiversity conservation, and ecological research. However, existing studies on TS classification predominantly rely on high-cost and processing-intensive hyperspectral data, which limits practical applications on large scales. In this work, we focus on investigating the potential of cost-effective unmanned aerial vehicle (UAV) RGB images for TS classification in heterogeneous forests and propose a method that fully leverages the rich spatial, semantic, and visible spectral information of UAV RGB images. We propose an RSVMamba model, which incorporates improved visual state-space (VSS) blocks and an AutoDownsampling module to enhance accuracy and stability while paying particular attention to small objects in sparse spatial locations. The model achieves linear computational complexity while retaining the global receptive field, making it particularly suitable for processing high spatial-resolution images. Additionally, we collected UAV RGB images covering 40 km^2 of subtropical forest in southern China. A meticulous evaluation of this data shows that our method achieves an overall accuracy (OA) of 84.28% for eight TS, dead trees, and other broadleaves. We verify the superiority of our method through a series of comparative experiments on the collected and benchmark datasets. Our results affirm the usefulness of single-temporal UAV RGB images for TS classification in heterogeneous forest environments. Furthermore, the proposed method bridges the gap between data accessibility and precision in TS classification, broadening the boundaries of single-temporal UAV RGB images for practical forestry applications and providing a more cost-effective and time-flexible solution for this problem.**

***Index Terms*— Deep learning, tree species (TS) classification, unmanned aerial vehicle (UAV) remote sensing.**

Received 13 August 2024; revised 23 November 2024; accepted 5 January 2025. Date of publication 23 January 2025; date of current version 4 February 2025. This work was supported in part by the Fundamental Research Funds for Natural Science Foundation of Heilongjiang for Distinguished Young Scientists under Grant JQ2023F002, in part by the Fundamental Research Funds for the Central Universities under Grant 2572020AW48 and Grant 2572023CT16, and in part by the China Scholarship Council. *(Corresponding authors: Weipeng Jing; Chao Li.)*

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Digital Object Identifier 10.1109/TGRS.2025.3533019

I. Introduction

TIMELY and accurate tree species (TS) classification is critical for offering comprehensive data support for biodiversity conservation, sustainable forest management, environmental monitoring, and urban planning [1]. Recently, unmanned aerial vehicle (UAV)-based remote sensing has become an important data source for this task. It allows for a more flexible and cost-effective acquisition of high-resolution remote sensing images compared to other aerial and spaceborne sensors [2]. Meanwhile, deep learning methods have demonstrated excellent performance in remote sensing image classification by effectively learning visual representations [3], [4]. Therefore, combining the operational convenience of UAVs with the prowess of deep learning visual models promises great potential for effective forest TS classification.

In recent years, numerous studies have been conducted in this direction [5]. Nevertheless, most of these studies rely on hyperspectral or multisource remote sensing data (e.g., hyperspectral data combined with LiDAR data) [6], which are costly and computationally intensive, limiting the feasibility for large-scale applications. Therefore, exploring feasible and cost-effective alternatives to hyperspectral data is of great practical significance.

Compared to sophisticated sensors, RGB (standard color) cameras deployed on unmanned aerial systems are cost-effective and easy to use. Moreover, they require minimal data preprocessing and can provide images with very high spatial resolution, enabling the capturing of detailed surface features [7]. These features include the shape, size, and color of tree canopies, all of which are highly informative for TS classification [8]. However, due to the limited spectral information of RGB data, it is difficult to distinguish TS with similar spectral characteristics, especially for forests with a high diversity of TS. Consequently, existing TS classification studies that only rely on RGB data generally classify fewer species and often leverage multitemporal RGB data. Furthermore, there is a lack of exploration of advanced deep visual models specifically designed for TS classification tasks using UAV RGB data, limiting their applicability for complex and real-time scenarios.

To this end, this study focuses on utilizing single-temporal UAV RGB images to achieve efficient forest TS classification. This approach reduces data acquisition costs and computational requirements while enhancing temporal flexibility,

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making it beneficial for practical applications. We aim to explore an advanced deep learning model to fully utilize the rich spatial, semantic, and visible spectral information present in UAV RGB data, bridging the gap between data accessibility and accuracy in TS classification tasks and providing a more cost-effective and time-flexible solution for the problem.

In the vision domain, vision Mamba (VMamba) [9] has recently emerged as an effective approach for global visual context modeling without the need for computationally expensive Transformer-based attention mechanism [10]. In this work, we propose RSVMamba, an approach inspired by Liu et al. [9] that is tailored to remote sensing images for the task of TS classification. Our approach adopts the second-order Runge–Kutta method (RK2) [11] for the higher-order discretization of the underlying state space, which is more suited to the task requiring high precision. This adoption enables cumulative error reduction by estimating the intermediate state in each time step, accurately reflecting the continuous-time dynamics of the system and improving the accuracy and stability. For the task at hand, there may be significant differences in population density among different TS within a forest. Some TS may form dense pure forests, while others may disperse or mix with other species, creating interspersed communities. This results in an uneven distribution and prediction bias, where the prediction accuracy of sparsely distributed TS tends to be lower. TS scattered within large tracts of pure forest are also often overlooked by prediction models, which has a negative impact on the performance and practical application of TS classification models. To address this, we additionally introduce an AutoDownsampling module in our approach that adaptively adjusts the downsampling stride to focus on small objects represented in sparse spatial locations.

Our approach particularly focuses on the characteristics of UAV RGB images and the challenges faced in TS classification using single-temporal images in complex forest environments. To evaluate our approach, we also introduce a UAV RGB image dataset specifically designed for TS classification. The effectiveness of the proposed approach is verified by the evaluation of this dataset and existing benchmark International Society for Photogrammetry and Remote Sensing (ISPRS) datasets [12], [13], while comparing performance with the state-of-the-art approaches. The specific contributions of this article are as follows.

1) It proposes the RSVMamba approach for TS classification using single-temporal UAV RGB images. Our method adopts a higher-order discretization for the state-space model and employs improved visual state-space (VSS) blocks. Moreover, it leverages a proposed AutoDownsampling module for high accuracy.
2) It introduces a UAV RGB image dataset collected over a 40-km^2 area in Qimen County, Huangshan City, Anhui Province, China. This dataset provides annotations for eight TS, dead trees, and other broadleaves.
3) It presents extensive analysis and results for the proposed, ISPRS Vaihingen [12], and ISPRS Potsdam [13] datasets. The results validate that single-temporal UAV RGB modality has the potential to achieve acceptable levels of multi-TS classification accuracy in complex heterogeneous forests. This portends more cost-effective and time-efficient technology for forest TS classification.

II. Related Work

A. Tree Species Classification Using UAV Hyperspectral Data

In the study of TS classification using UAV remote sensing images, hyperspectral data is the mainstream modality [14]. Among the methods leveraging this modality, [15], [16], and [17] have obtained impressive classification results using only hyperspectral data, with accuracy typically surpassing 85%. Additionally, [18], [19], [20], and [21] have achieved the accurate classification of various TS using deep learning models based on multisource remote sensing data including hyperspectral, LiDAR, and RGB data. These methods often achieve overall accuracies (OAs) exceeding 90%. In general, using UAV hyperspectral data is sufficient to achieve fine and accurate classification of multiple TS in complex forest environments, and the assistance of LiDAR and RGB data further improves performance [14].

B. Tree Species Classification Using UAV RGB Data

Existing research [22] indicates that when using the texture, vertical structure, and spectral features extracted from UAV RGB, LiDAR, and hyperspectral data for TS classification, the performance based on RGB data is typically the lowest. Consequently, studies focusing on TS classification using only RGB data are relatively limited. Among related studies that incorporate deep learning methods [23], [24], [25], [26], [27], the classification accuracy varies considerably due to the differences in the number of TS and the classification levels in the datasets. The broad approximate range of the performance remains within 75%–90%, and the number of TS is often limited, which is difficult to apply to complex forest environments with rich TS. Among these methods, Natesan et al. [23] explored the differences between multitemporal and single-temporal UAV RGB images in identifying two types of pine trees using residual networks (ResNets) [28]. The results indicated that the classification accuracy of multitemporal data was nearly 30% higher than that of single-temporal data. Pierdicca et al. [26] explored the application of different deep learning architectures, including ResNet, densely connected convolutional networks (DenseNet), InceptionV3, and Vision Transformer, for classifying four TS based on UAV RGB images. The experimental results showed that the classification accuracy of the Vision Transformer significantly exceeds that of the other models. This indicates that multitemporal data and different deep learning methods have a great impact on the performance of the TS classification task. However, the use of multitemporal data limits real-time applicability in practical scenarios, and these deep learning methods do not account for the specific challenges of TS classification tasks, such as interspecies similarity and sparsely distributed TS. Therefore, exploring effective TS classification methods based on low-cost single-temporal UAV RGB data is of great significance.

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C. Tree Species Classification Methods

In early TS classification studies based on remote sensing, traditional methods commonly used parametric classifiers such as maximum likelihood and linear discriminant analysis [29], which often rely on strict assumptions about the data distribution. To overcome this limitation, machine learning models such as random forest and support vector machines were used to classify multiple TS in combination with different UAV remote sensing data [30]. However, the performance of these models generally relies on appropriate feature selection. In recent years, deep learning methods have been extensively employed in TS classification [6], [31], [32]. Deep learning methods can automatically learn feature representations from the original data, thereby reducing the dependence on manual feature selection while exhibiting strong adaptability. However, most current studies on TS classification still use classical convolutional neural network (CNN)-based models such as Inception [33], ResNet [28], and DenseNet [34]. The exploration of the more recent deep neural models with better performance and stronger feature representation capabilities is still limited. Hence, an important contribution of our work is to adapt one of the most advanced deep learning-based visual modeling techniques for TS classification.

D. Summary

In summary, TS classification using UAV remote sensing data has achieved significant progress. In terms of data sources, existing studies on TS classification predominantly rely on hyperspectral data due to its rich spectral information [14] and often combined with LiDAR or RGB data, which is sufficient to achieve high-precision classification of multiple TS. However, the high cost and complexity of hyperspectral data acquisition limit its scalability for large-scale applications. In contrast, UAV RGB data offer a low-cost and accessible alternative, but there are relatively limited studies in this area, most of which rely on multitemporal data, restricting temporal flexibility in practical applications. Methods based on single-temporal RGB data remain underexplored, particularly in classifying multiple TS in complex forest environments, which still poses significant challenges.

In terms of classification methods, deep learning methods have been widely applied to TS classification [6], [31], [32]. However, most studies still employ conventional CNN models, lacking exploration of more advanced deep vision models, especially those specifically tailored for TS classification tasks. Therefore, this study focused on exploring high-performance methods specifically adapted for TS classification tasks based on single-temporal UAV RGB data, providing a cost-effective alternative to hyperspectral data.

III. Research Area and Data

A. Research Area

The research area is located in Qimen County, Huangshan City, Anhui Province, China (29°35'61"–30°08'30"N, 117°11'53"–117°57'11"E) [see Fig. 1(a)]. This region has a subtropical monsoon climate with hot and humid summers, mild and dry winters, and obvious seasonal variations in spring and autumn. The precipitation is relatively evenly distributed throughout the year, and the average annual temperature is between 11.0 °C and 22.0 °C, with an extreme maximum temperature of 38.0 °C and a minimum temperature of −8.0 °C. Due to its specific geographical location and climatic conditions, Qimen County is suitable for the growth of various plants and has rich natural landscapes and biodiversity. It is also a major county for forest resources in Anhui Province, with a forest land area of 203 300 ha, a total standing forest stock of 11.75 million cubic meters, and a forest coverage rate of 88.64%. The total area of the research area is about 40 km^2, where plantation forests account for about 70%, and natural forests account for about 30%. The forest environment is complex and varied, with a highly closed canopy and complex topography. The forest types are diverse, mainly including coniferous forests, broadleaved forests, mixed coniferous broad-leaved forests, and bamboo forests, among which coniferous forest is dominated by *Cunninghamia lanceolata*, *Pinus elliottii*, while the broadleaf forest is dominated by *Quercus spp.*, *Liquidambar formosana*, and *Eucalyptus globulus*, and bamboo forest is mainly dominated by *Phyllostachys edulis*.

TABLE I
UAV Flight Parameter Setting Information

Aircraft	DJI Matrice 300 RTK
RGB camera	Zenmuse P1
Resolution	45 megapixels
Ground Sample Distance	6.15 cm/pixel
Spatial resolution	0.05 m
Maximum flight altitude	490 m above ground level
Flight speed	13 m/s
Forward overlap rate	80%
Side overlap rate	70%
Photo mode	Equidistant shooting

B. Collection of the Tree Species Dataset

Based on the detailed information provided by the forest resources investigation data in Qimen County in 2020, including forest coverage rate, forest types, TS composition, forest age structure, and forest land quality, we identified the scope of the research area as shown in Fig. 1(b), which has a diverse and rich distribution of TS. High-spatial-resolution RGB images were collected using a DJI Matrice 300 RTK equipped with the Zenmuse P1 high-resolution RGB camera, which has 45 million pixels. Data collection was conducted under windless, clear weather conditions on October 18th and 19th, 2022, the details of RGB data collection are presented in Table I.

The distribution of the sample plots is shown in Fig. 1(d). Due to the presence of closed roads and inaccessible areas within the region, the sample plots are not all distributed within the research area but are evenly distributed within Qimen County, where the research area is located. According to the forest resources inventory data, the environmental conditions

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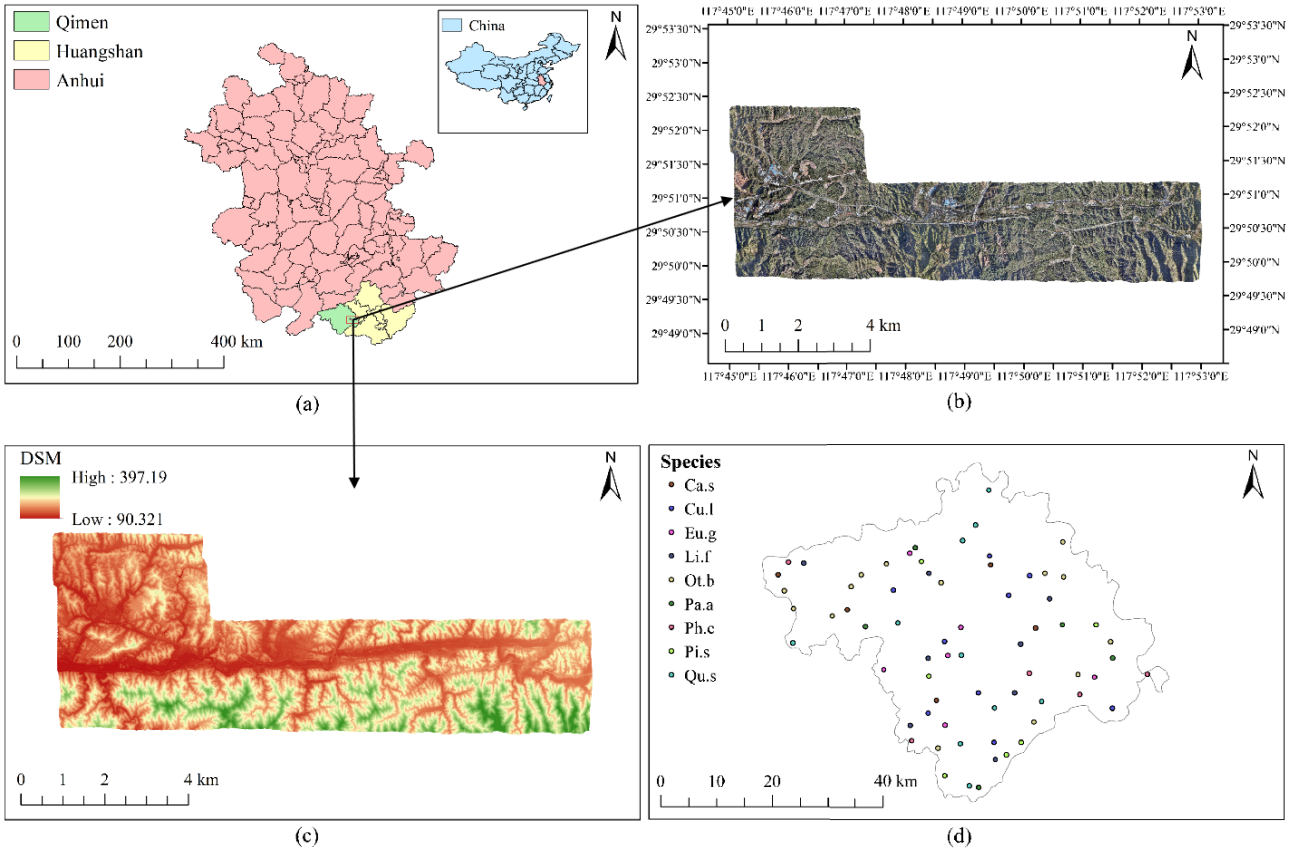


Fig. 1. Overview of the research area. (a) Location of the research area. (b) Combination image of UAV RGB data. (c) DSM of the research area. (d) Distribution of the sample plots.

TABLE II
DOMINANT TS AND QUANTITY OF SAMPLE PLOTS

Tree Species	Species Merged	Shorthand	Sub-compartment	Sampling Plot
Pinus elliottii	*Pinus elliottii*	Pi.s	130	6
Cunninghamia lanceolata	*Cunninghamia lanceolata*	Cu.l	555	9
Quercus variabilis, *Quercus aliena*	*Quercus spp.*	Qu.s	178	11
Liquidambar formosana	*Liquidambar formosana*	Li.f	105	8
Eucalyptus globulus	*Eucalyptus globulus*	Eu.g	80	6
Phyllostachys edulis	*Phyllostachys edulis*	Ph.e	60	5
Camellia sinensis	*Camellia sinensis*	Ca.s	435	5
Paulownia	*Paulownia*	Pa.a	28	5
Melia azedarach, *Catalpa*, *Alnus*, *Schima*, *Populus*, *Sassafras*, *Cinnamomum camphora*, *Santalum*, *Jatropha*, *Pteroceltis*	Other broadleaf	Ot.b	243	14

[1] The fourth column is the number of sub-compartments of dominant tree species in the research area.

and TS distribution in the research area and Qimen County are relatively consistent. The TS included in the sample plots covers the main TS in the area as well as all the species involved in this study. Therefore, the sample plot data are representative of the research area. The sampling survey was carried out in 69 sample plots with a size of 25.8 × 25.82 m in Qimen County, which mainly included eight dominant TS, such as *Pinus elliottii*, *Cunninghamia lanceolata*, *Quercus spp.*, *Liquidambar formosana*, *Eucalyptus globulus*, *Phyllostachys edulis*, *Paulownia*, and *Camellia sinensis*. The investigation factors included the latitude and longitude of the plot center and corner points, topography, soil, vegetation type, canopy density, dominant TS, TS type, average tree height, and so on. Here, we only list the survey factors related to this study, and the corresponding dominant TS, the number of subcompartments, and sample plots are shown in Table II.

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TABLE III
DISTRIBUTION OF DIFFERENT CATEGORIES IN THE PROPOSED TS DATASET

Tree Species	Total Image Count	Total Pixel Count	Class-wise Distribution (%)	Pixel-wise Distribution (%)
Pinus elliottii	86	7.29×10^7	43.00	3.33
Cunninghamia lanceolata	154	3.36×10^8	77.00	15.34
Quercus spp.	67	2.40×10^8	33.50	10.95
Liquidambar formosana	170	1.65×10^8	85.00	7.53
Eucalyptus globulus	126	6.01×10^7	63.00	2.74
Phyllostachys edulis	115	1.52×10^8	57.50	6.94
Camellia sinensis	100	5.89×10^7	50.00	2.69
Paulownia	51	2.54×10^7	25.50	1.60
Other broadleaf	183	6.56×10^8	91.50	29.94
Dead tree	52	1.03×10^7	26.00	0.47
Background	194	4.15×10^8	97.00	18.94

[1] The above statistics are based on 200 pairs of original images and labels.



Fig. 2. Example of RGB remote sensing image, DSM, and label.

C. Data Processing

Considering the general composition and distribution of TS in the research area, as well as the ecological benefits of different TS, our research mainly focuses on the classification of eight major TS, including *Pinus elliottii*, *Cunninghamia lanceolata*, *Quercus spp.*, *Liquidambar formosana*, *Eucalyptus globulus*, *Phyllostachys edulis*, *Vernicia fordii*, *Camellia sinensis*, Dead tree, and Other broadleaf. The TS included in Other broadleaf are shown in Table II. Due to the complex distribution and high interspecies similarity of these broadleaf trees, it is challenging to accurately label and identify them in remote sensing images. Therefore, they are all categorized as a single group in this study, with no further detailed differentiation. In addition, canopies that could not be classified due to the fuzzy image areas, as well as TS not included in this study and other nonforest land such as buildings, farmland, and roads, were classified as category "Background" and excluded from further analysis. We utilized ArcGIS 10.6 to mosaic and crop the original images. To ensure the mosaic quality, 80% forward overlap and 70% side overlap were maintained during data acquisition, along with consistent collection conditions. The high overlap rates provided sufficient information for alignment and registration, ensuring ample common feature points to minimize misalignments, stitching gaps, and boundary effects. Consistent collection conditions also reduced pixel discrepancies and geometric errors, thereby enhancing the data continuity of the mosaic. Additionally, edge smoothing and geometric corrections were applied to further ensure mosaic quality. The mosaicked images were subsequently cropped into 1024 $\times$ 1024 patches, excluding those containing large nonforest areas. The UAV RGB images were manually annotated based on the field survey data and forest resources investigation data. Delineation and category assignment were cross-checked by at least one additional interpreter to ensure accuracy. Fig. 2 shows examples of original RGB remote sensing images, digital surface model (DSM) data, and labels of the research area.

We sampled 200 pairs of original images and labeled images of size 4096 $\times$ 2730 from the TS dataset. The available data is randomly divided into a training set and a testing set with a ratio of 3:1. We provide a brief statistical analysis of the distribution of different categories within the 200 samples of our TS dataset in Table III.

IV. METHOD

In this section, we introduce RSVMamba, a VMamba [9] inspired neural model to process remote sensing UAV RGB images for effective TS classification. The underlying network architecture of RSVMamba is shown in Fig. 3. The proposed approach exploits two innovative concepts in terms of improved VSS blocks and an AutoDownsampling module. We first discuss the network architecture and then focus on these components.

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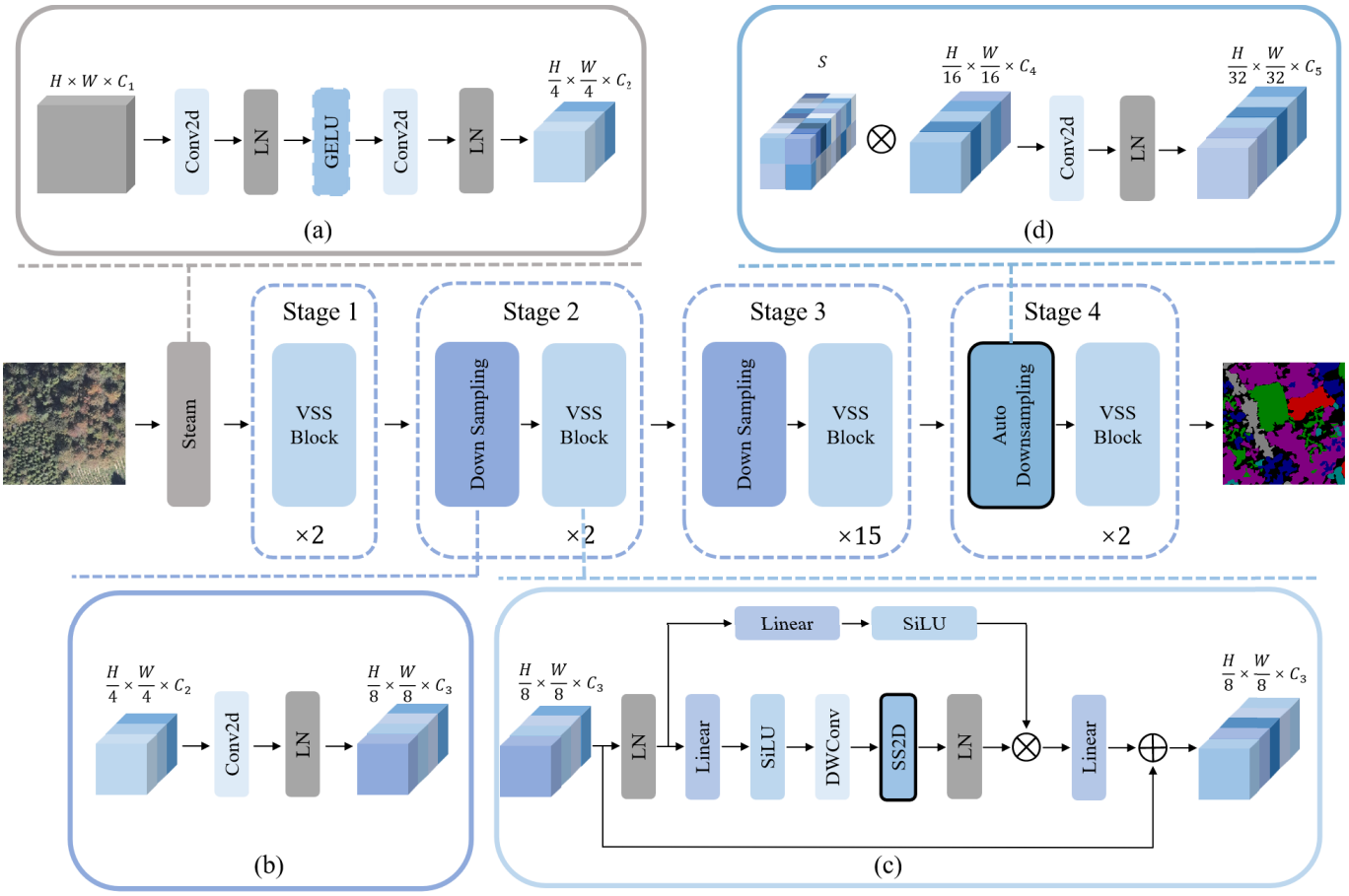


Fig. 3. Overall architecture of the proposed RSVMamba. The Steam module converts the input image into a sequence of patches, which are processed through a series of stacked VSS blocks and Downsampling modules. (a)–(d) Details of the Steam module, Downsampling module, VSS block, and AutoDownsampling module, respectively. Improvements to the model are marked by solid black boxes. In our experiments, $H = 1024$, $W = 1024$, $C_1 = 3$, $C_2 = 128$, $C_3 = 256$, $C_4 = 512$, and $C_5 = 1024$. The S in (d) represents the score matrix composed of the importance score s_i.

A. Network Architecture

The overall architecture of the proposed RSVMamba is illustrated in Fig. 3. An image with a dimension of $H \times W \times C_1$ is processed through a stem module, the structure of the module is illustrated in Fig. 3(a). The processing preserves the 2-D structure of the original image, resulting in a feature map of dimension $(H/4) \times (W/4) \times C_2$. Then, four stages are designed to further process the acquired feature maps. Stage 1 consists of two stacked VSS blocks, and Stage 2 and Stage 3 consist of a Downsampling module and several stacked VSS blocks. The Downsampling modules use a conventional 2-D convolutional layer followed by layer normalization (LN) to reduce the feature maps to half along the spatial dimensions. Eventually, at the output of Stage 3, we receive $(H/16) \times (W/16) \times C_4$ maps. Stage 4 is composed of an AutoDownsampling module and two stacked VSS blocks. The former weights the output of Stage 3 with the importance score matrix and reduces the feature size along the spatial dimensions by half. This is followed by two stacked VSS blocks. Eventually, classification results are obtained through a softmax operation.

B. Improved Visual State Space

A detailed structure of our VSS block is illustrated in Fig. 3(c). As shown, the output of the preceding module is processed by the VSS block under two branches after LN. Branch 1 includes a linear layer and a sigmoid linear unit (SiLU) activation operation. Branch 2 includes a linear layer, a SiLU activation operation, a depth-wise (DW) convolution operation, the 2-D selective scan (SS2D) module [9], and LN. An element-wise multiplication operation is performed on the output of Branch 1 and Branch 2, then an addition operation is performed on the result and the original input to get the final output of the VSS block. In the VSS block, the SS2D module is composed of the cross-scan module (CSM) and an S6 block as shown in Fig. 4. The CSM expands the image patches into sequences along the four directions of rows and columns and then traverses the spatial domain along the four sequences to integrate pixel information from different directions. The input sequences are then processed by the S6 module through a series of linear transformations and discretization processes, which retain the linear complexity of the selective scanning mechanism while preserving the global receptive field. Finally, all sequences are merged into one complete image.

As the core element of SS2D, the state-space model of the continuous domain is usually constructed as linear ordinary differential equations [9], as shown in the following equation:

$$h'(t) = Ah(t) + Bx(t)$$
$$y(t) = Ch(t) + Dx(t) \quad (1)$$

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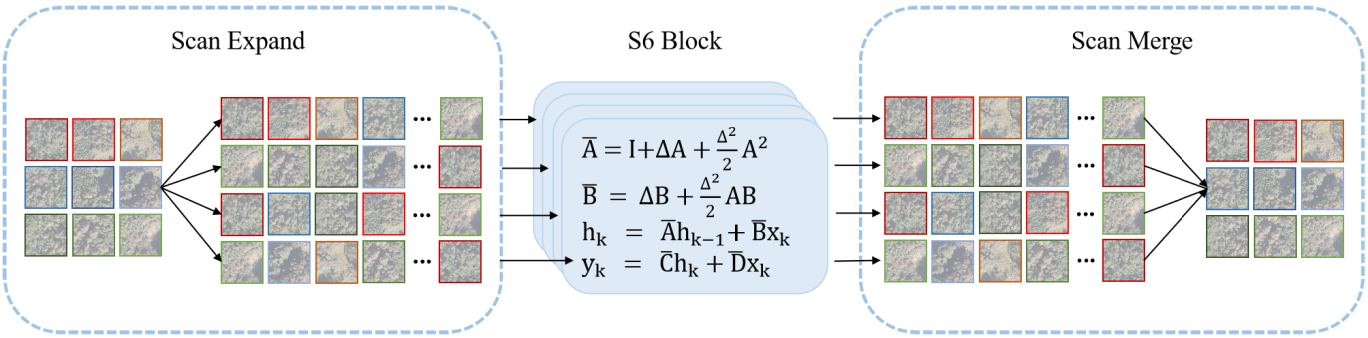


Fig. 4. Structure of the SS2D on an image. The scan expand module scans and expands the image into four sequences along the four directions of row and column, the four sequences are processed separately in the S6 block, and finally, the scan merge module combines all the output sequences into one feature map.

where A denotes the state transition matrix, B denotes the control input matrix, C denotes the output matrix, and D denotes the direct transmission matrix. $h(t)$ denotes the state vector representing the internal state of the system at time t and $h'(t)$ denotes the derivative of the state vector representing the changing trend of the state variable. $x(t)$ and $y(t)$ denote the inputs and outputs, respectively.

The processing of information by computers is based on discrete time steps, which require the transformation of continuous system state equations into discrete equations for implementation in the computational framework. Additionally, deep learning frameworks are generally developed to deal with discretized data, making the discretization process indispensable. Considering that the RK2 [11] provides improved stability and accuracy compared to lower-order discretization methods such as the Zero-Order Hold Rule, which only approximate the state of the sampling points and ignores the dynamic changes within the interval, thereby introducing local errors that accumulate as the number of sampling points increases. In addition, compared to higher-order discretization methods, such as the fourth-order Runge–Kutta method, require multiple derivative calculations within each time step. As the number of sampling points increases, this significantly escalates the computational burden. In contrast, RK2 has a lower computational cost, as the number of sampling points increases, the computational burden of RK2 grows moderately while maintaining high accuracy, achieving an optimal balance between accuracy and computational efficiency, making it suitable for our application scenarios. Therefore, we employ RK2 to discretize the continuous state-space system in our approach. RK2 calculates the intermediate state within each time step and approximates the continuous-time dynamics of the system better. This higher-order approximation reduces the local truncation error within each time step, thereby decreasing the cumulative errors and numerical instability, which improves the overall precision of the discretization. For the input $x(k) \in \mathbb{R}^L$, the sampled vector within a signal flow of length L, and assuming $x(k)$ and $h(k)$ are smooth enough in the interval, the ordinary differential equation is discretized by RK2 as follows:

$$\begin{aligned} k_1 &= Ah(t_k) + Bx(t_k) \\ k_2 &= A(h(t_k) + \Delta k_1) + Bx(t_k) \\ h(t_{k+1}) &= h(t_k) + \frac{\Delta}{2}(k_1 + k_2). \end{aligned} \tag{2}$$

In the above, Δ denotes the time step interval, and replacing the time t_k with the discrete time step k, the above process is expressed as a discretized form of the state-space model

$$\begin{aligned} h_k &= \overline{A}h_{k-1} + \overline{B}x_k \\ y_k &= \overline{C}h_k + \overline{D}x_k \\ \overline{A} &= I + \Delta A + \frac{\Delta^2}{2}A^2 \\ \overline{B} &= \Delta B + \frac{\Delta^2}{2}AB \\ \overline{C} &= C \\ \overline{D} &= D. \end{aligned} \tag{3}$$

In the above, $\overline{A}$ denotes the discrete system matrix, $\overline{B}$ denotes the discrete input matrix, $\overline{C}$ denotes the output matrix, and $\overline{D}$ denotes the direct transmission matrix. $\overline{A} \in \mathbb{R}^{N\times N}$, $\overline{B} \in \mathbb{R}^{N\times L}$, $\overline{C} \in \mathbb{R}^{P\times N}$, $\overline{D} \in \mathbb{R}^{P\times L}$ for a state size N, an input size L, and an output size P.

C. AutoDownsampling

We introduce an AutoDownsampling module to improve the traditional grid downsampling for the feature maps. We evaluate the importance of each token in the data according to the extracted features and dynamically adjust the sampling stride. First, the importance of each token in the feature map is evaluated by calculating an importance score. We aim to measure the importance of various aspects such as data features and spatial structure information. Therefore, for any token $t_i = (x_i, y_i)$, the importance score is calculated as $s_i = \alpha p_i + \beta l_i$. This is the weighted sum of the learning score l_i of the feature and the grid prior score p_i, where α and β are the hyperparameters that can be adjusted according to the task. $l_i = \sigma(f_i)$ is the feature-based importance score obtained through the sigmoid function $\sigma(\cdot)$.

The grid priors provide the spatial relationships and structural information of the original data, improving the ability to distinguish similar tokens in different locations. This promotes the model to execute appropriate uniform strides in local textureless areas, providing a structured framework for downsampling and guiding decision-making during sampling. Meanwhile, we also consider the local density of each token based on traditional grid sampling and dynamically allocate the sampling stride. Therefore, we introduce a grid prior p_i.

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Specifically, for each token t_i, we calculate the average distance to its K nearest neighbors and assign a downsampling stride to it. The average distance reflects the overall density of the local area, and reduces the impact of noise and outliers, thus enhancing robustness. The calculation is as follows:

$$d_i = 2^{\lceil \log_2 \frac{\sum_1^K \min_j \|t_i - t_j\|_1}{K} \rceil}. \tag{4}$$

For each token $t_i = (x_i, y_i)$, if $x_i \bmod 2d_i = 0 \wedge y_i \bmod 2d_i = 0$, $p_i = 1$, else $p_i = 0$. For example, if the ceiling of the average distance to the K nearest neighbors of t_i is 1, then the assigned stride d_i is $2^0 = 1$, and each token is alternately assigned $p_i = 1$, and $p_i = 0$. If the ceiling of the average distance is 3, then the assigned stride d_i is $2^2 = 4$, and every four tokens are assigned $p_i = 1$, with the rest assigned $p_i = 0$. So, the tokens not selected are influenced only by the learned score l_i and not by the grid prior p_i, which ensures that only important tokens that conform to the grid structure are retained, while unimportant or nonconforming points are disregarded. Finally, the importance score s_i of each token constitutes the weight matrix S, and performs element-wise multiplication with the original feature map. The output of the AutoDownsampling module is obtained after a convolution operation and LN as shown in Fig. 3(d). Moreover, this operation is only performed for the last downsampling, so the additional computation does not increase significantly.

V. Experiment and Discussion

We perform extensive evaluations on the proposed TS dataset and existing ISPRS Vaihingen [12] and Potsdam datasets [13]. Details of the TS dataset are already provided in Section III. Therefore, we do not repeat them here. We briefly introduce the ISPRS datasets, evaluation metrics, implementation details, and the experimental setup. Our main focus here is on the results of the experiments and the subsequent analysis.

A. ISPRS Datasets

To validate the generalization capabilities of the proposed model, we conducted a series of experiments on the ISPRS Vaihingen [12] and Potsdam datasets [13], which consist of the true ortho photo (TOP), ground-truth images, and DSM. The ISPRS Vaihingen dataset comprises 33 TOP images with an average size of 2064 × 2494 pixels and the ISPRS Potsdam dataset comprises 38 TOP images with a size of 6000 × 6000 pixels. Both datasets involve six categories: "Impervious surface," "Building," "Low vegetation," "Tree," "Car," and "Clutter." Due to the incomplete labeling of the category "Clutter," it was excluded from the study. In our experiments, we used the TOP images and ground-truth images, which were divided into a training set and a testing set in a ratio of 2:1. Additionally, to focus on the classification objectives related to this study, we further processed the ISPRS Vaihingen dataset, merging the categories "Impervious surface," "Building," "Car," and "Clutter" into the category "Background" to construct a three-categories dataset and assess the classification accuracy of the categories "Tree" and "Low vegetation." The cropping and division methods are consistent with the original ISPRS Vaihingen dataset.

TABLE IV
Experimental Environment and Parameter Setting

Platform	PyTorch
GPU	NVIDIA A100
CUDA Version	12.1
Optimizer	AdamW
Learning rate	6×10^{-5}
Weight decay	0.01
Iterations	80,000
Batch size	16
input size	1024×1024

B. Evaluation Metrics

At the pixel level, the task of TS classification can also be considered an image segmentation task. Hence, its evaluation requires metrics commonly employed in the segmentation task. We perform our quantitative evaluation using three common semantic segmentation metrics, namely, mean intersection over union (mIoU), mean F1 score (AF), and OA.

C. Implementation Details

We implement all the experiments on the PyTorch platform with the NVIDIA A100 GPU. Since our approach is originally inspired by VMamba [9], we also take guidance on hyperparameter settings for our training from [9]. Our method uses the AdamW optimizer, setting the learning rate to 6×10^{-5} and the weight decay to 0.01. The training process spanned 80 000 iterations in total, with a batch size of 16 and an input size of 1024 × 1024 for both the ISPRS datasets and the TS dataset. The specific parameter settings are shown in Table IV.

D. Experiments on the TS Dataset

To evaluate the performance of the proposed RSVMamba, we conducted comparative evaluations using CNN-based models, Transformer-based models, and Mamba-based models, which include fully convolutional network (FCN) [35], UNet [36], ResNet50 [28], Deeplab v3+ [37], TransUNet [38], densely connected swin transformer (DCSwin) [39], attentive bilateral contextual network (ABCNet) [40], UNet-like transformer (Unetformer) [41], CNN and multiscale transformer fusion network (CMTFNet) [42], and VMamba [9].

Table V presents the results of the aforementioned models on the TS dataset. It indicates that for the three standard evaluation metrics, mIoU, AF, and OA, the proposed model achieves scores of 71.12%, 81.45%, and 84.28%, respectively. Among the other evaluated models, VMamba demonstrates the most promising segmentation accuracy. However, for the proposed technique, the mIoU, AF, and OA values improve by an absolute 0.94%, 1.38%, and 1.05% over VMamba, which is significant. The results indicate that the proposed model is superior to the existing models in segmentation accuracy for most categories, and the overall performance surpasses the other methods. This demonstrates the effectiveness of the proposed model in extracting category-specific information. Especially for categories with sparse distribution, such as

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TABLE V
EXPERIMENTAL RESULTS ON THE TS DATASET, WITH THE BEST RESULTS INDICATED IN BOLD (EXCEPT FOR RSVMAMBA+DSM)

Method	Pi.s	Cu.l	Qu.s	Li.f	Eu.g	Ph.e	Ca.s	Pa.a	Ot.b	De.t	Ba.d	mIoU(%)	AF(%)	OA(%)
FCN [35]	56.32	65.74	44.81	64.24	72.36	69.72	70.47	54.55	49.19	38.92	72.63	59.91	69.57	70.45
UNet [36]	59.11	68.90	45.53	66.86	75.02	73.24	72.33	56.75	52.69	40.18	75.47	62.37	72.26	73.34
ResNet50 [28]	59.18	70.37	47.61	67.49	77.22	76.50	73.73	59.95	56.86	40.61	78.34	64.35	76.47	75.99
Deeplab v3+ [37]	58.63	68.95	46.72	67.51	78.46	78.99	74.80	60.87	56.14	39.01	79.27	64.49	76.61	76.93
TransUNet [38]	60.12	70.45	49.64	70.78	79.82	78.97	76.59	63.36	57.18	42.21	84.93	66.73	78.09	79.93
DCSwin [39]	61.83	72.06	49.12	72.75	80.68	80.37	79.53	64.49	58.94	44.86	84.46	68.10	80.08	81.92
ABCNet [40]	63.61	73.09	50.34	72.87	82.15	80.53	78.26	64.47	59.70	45.56	85.38	68.72	79.82	81.61
CMTFNet [42]	65.77	74.89	50.83	**74.98**	83.81	81.34	79.06	65.29	60.52	44.65	86.14	69.75	80.00	82.31
UNetFormer [41]	64.71	74.16	50.65	73.19	83.27	80.93	79.42	65.54	60.06	46.38	86.79	69.55	79.48	82.29
VMamba [9]	66.79	75.17	50.83	73.94	85.37	80.85	79.72	66.48	60.86	**46.41**	85.53	70.18	80.07	83.23
VMamba+AD	67.01	75.28	50.87	74.21	85.54	81.54	79.96	67.27	60.73	46.32	85.61	70.39	80.42	83.54
VMamba+RK2	67.47	75.86	51.49	74.68	85.83	81.59	80.54	67.53	61.69	46.31	86.74	70.88	81.11	84.18
RSVMamba (ours)	**67.69**	**76.08**	**51.97**	74.76	**86.28**	**82.32**	**80.74**	**67.75**	**61.67**	46.27	**86.81**	**71.12**	**81.45**	**84.28**
RSVMamba+DSM	69.12	76.97	52.04	75.36	87.09	83.72	82.63	67.77	62.28	46.34	88.16	71.95	82.52	85.61

[1] VMamba+AD is the model in which the AutoDownsampling module is introduced in VMamba. VMamba+RK2 is the model in which only the RK2 discretization method is introduced to VMamba. RSVMamba+DSM is the model that introduced DSM data as an additional input. Both these methods make contributions to this work.

"Ph.e" and "Pa.a," there was a significant improvement in classification accuracy. Specifically, the mIoU of category "Ph.e" increased by 1.47%, and of category "Pa.a" by 1.27%. We believe this improvement is due to the use of a higher-order RK2 method to discretize the state-space model, which improved the overall classification accuracy. Additionally, the introduction of the AutoDownsampling module helps retain important feature information, preventing these small targets from being overlooked or misclassified, which plays a key role in improving the classification accuracy of categories "Ph.e" and "Pa.a."

In addition, we conducted experiments that incorporated the DSM data as an additional input to assess its impact on classification results. The results are shown in Table V, where the last row and the second-to-last row display the experimental results of UAV RGB combined with DSM data and of UAV RGB data alone, respectively. The introduction of DSM data increased mIoU, AF, and OA by 0.83%, 1.07%, and 1.33%, respectively, indicating its positive impact on TS classification. We believe that different TS exhibit varying heights, which are not prominently captured in RGB data, while DSM data provides 3-D structural information, such as height and canopy density. By combining DSM and RGB data, the model can better distinguish the TS with different heights and morphological characteristics. Consequently, the classification accuracy was significantly improved for "Ca.s," a low-growing shrub. Additionally, the classification accuracy of species with high height and prominent morphological characteristics, such as "Pi.s" and "Ph.e," also improved to a certain extent.

We note that our method did not achieve the best results in the categories "Li.f" and "De.t." However, our performance is very close to the best-performing methods. It is noteworthy that our remote sensing images were collected in mid-October when it was autumn in southern China. *Liquidambar formosana*, as a deciduous tree, entered its color-changing phase during this time, with its leaves turning red, orange, or yellow, making it easily confused with other color-changing deciduous TS. Similarly, "De.t" exhibits the lowest accuracy among all categories because some deciduous trees are in the defoliating period, leading to confusion between deciduous and dead trees. Furthermore, lighting and shadows also further adversely affect the classification accuracy of this category. We also note that the segmentation performances on the categories "Qu.s," "Pa.a," and "Ot.b" are generally not too high for all the models. We conjecture that the main reason for this is that the categories "Qu.s" and "Ot.b" are both genus-level categories that encompass multiple TS, as shown in Table II, which results in large intraclass variance. Additionally, Quercus itself, also being a broadleaf tree, has great similarities with other broadleaf trees in terms of color, texture, leaf morphology, and so on, which leads to confusion between these two categories and brings great challenges for accurate classification. The category "Pa.a" is less distributed in the study area and lacks sample abundance. Hence, it is likely mixed with other dominant species, making it difficult to accurately identify.

To visually display the differences in classification results of the aforementioned models, we visualized the predicted results, as shown in Figs. 5 and 6. It can be observed that the overall performance of RSVMamba surpasses that of other comparative methods, with clearer boundaries between different categories. Specifically, in the area with relatively uniform TS distribution and no intermixing as shown in Fig. 5, most models can accurately identify the species and their boundaries for TS with significant interclass differences. However, for TS that are less distinct and easily confused, such as "Qu.s" and "Ot.b," with blurred boundaries as shown by the red dashed box in Fig. 5, our method achieves better classification and clearer boundaries. In contrast, for the large homogeneous areas of the same TS shown in

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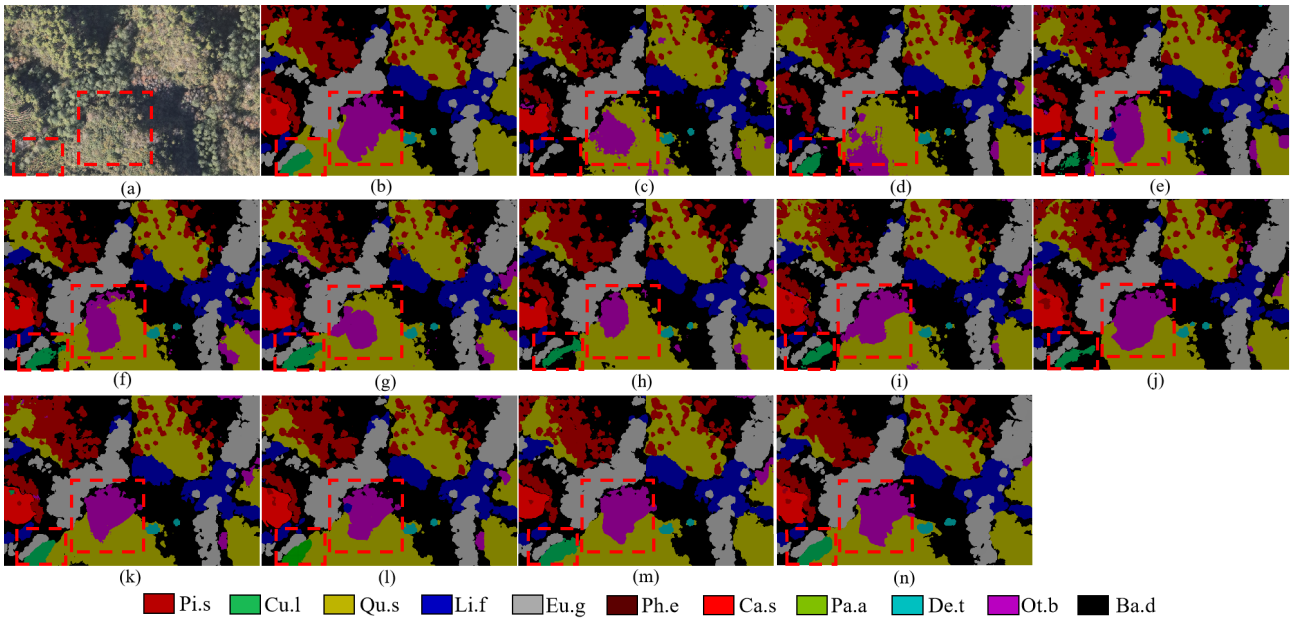


Fig. 5. Visualization results of the TS dataset. (a) Original image. (b) Ground truth. (c) FCN [35]. (d) UNet [36]. (e) ResNet50 [28]. (f) Deeplab v3+ [37]. (g) TransUNet [38]. (h) DCSwin [39]. (i) ABCNet [40]. (j) Unetformer [41]. (k) CMTFNet [42]. (l) VMamba [9]. (m) RSVMamba. (n) RSVMamba+DSM.

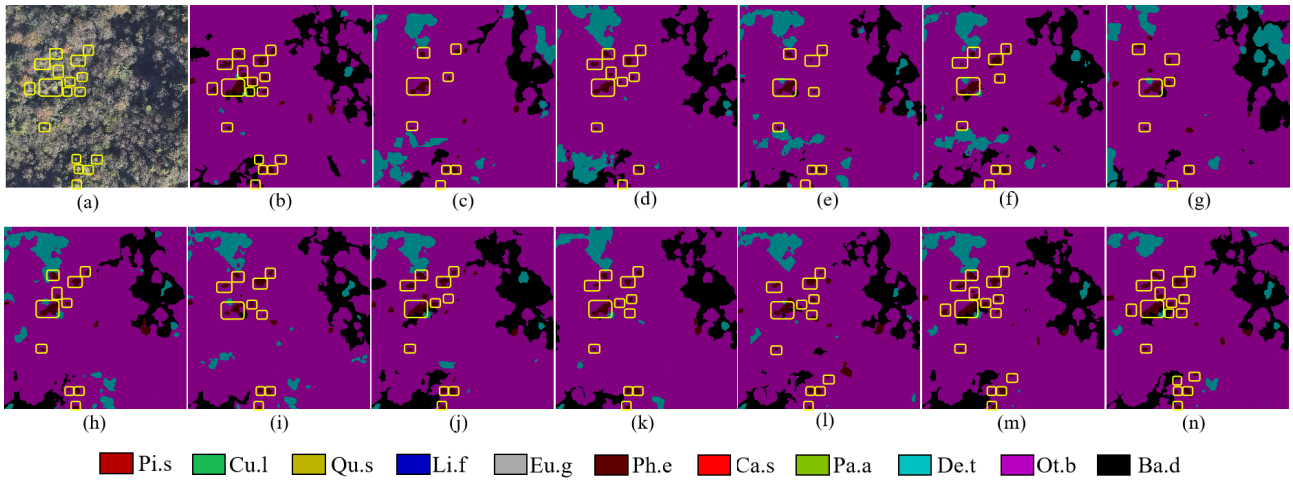


Fig. 6. Visualization results of the TS dataset. The yellow solid-line boxes in (a) and (b) highlight examples of the category "Ph.e," the yellow solid-line boxes in (c)–(m) indicate the parts accurately identified by the model compared to the examples in (b). (a) Original image. (b) Ground truth. (c) FCN [35]. (d) UNet [36]. (e) ResNet50 [28]. (f) Deeplab v3+ [37]. (g) TransUNet [38]. (h) DCSwin [39]. (i) ABCNet [40]. (j) Unetformer [41]. (k) CMTFNet [42]. (l) VMamba [9]. (m) RSVMamba. (n) RSVMamba+DSM.

Fig. 6, the classification results for "De.t" are generally less satisfactory, with considerable confusion with deciduous TS. For the sporadically distributed "Ph.e," it is also challenging to accurately and comprehensively identify them. We have marked some scattered *Phyllostachys edulis* in the original images and labels, as visible in the yellow solid boxes in Fig. 6(a) and (b). The yellow solid boxes in Fig. 6(c)–(n) indicate the *Phyllostachys edulis* correctly identified by all the methods in the corresponding positions. The results show that our method can more accurately recognize them, whereas other models tend to overlook such small targets.

E. Experiments on the ISPRS Dataset

To verify the generalization ability of the proposed model, we conducted a set of experiments on the public datasets: the ISPRS Vaihingen dataset and the ISPRS Potsdam dataset. The prediction results are presented in Tables VI and VII. It shows that for the standard evaluation metrics, mIoU, AF, and OA, the proposed model achieved results of 84.14%, 92.76%, and 92.13% on the ISPRS Vaihingen dataset, and 87.49%, 92.96%, and 91.87% on the ISPRS Potsdam dataset. The results indicate that the proposed model outperforms other methods in the segmentation accuracy of most categories. Although the accuracy improvement relative to VMamba is not too drastic, the overall performance improves and shows consistency across both datasets. It is observable that the segmentation accuracy for the categories "Low vegetation" and "Tree" is slightly lower than that of other categories for all the models. We believe this is mainly due to the small inter-class differences between these two categories, particularly in terms of texture and color similarities, which makes them prone to confusion. Additionally, their widespread distribution,

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TABLE VI
EXPERIMENTAL RESULTS ON THE ISPRS VAIHINGEN DATASET, WITH THE BEST RESULTS INDICATED IN BOLD

Method	Imp. surface	Building	Low veg.	Tree	Car	mIoU(%)	AF(%)	OA(%)
FCN [35]	80.35	84.35	66.78	73.86	64.92	74.05	82.67	84.12
UNet [36]	82.79	86.74	68.25	75.91	66.36	76.01	83.58	85.63
ResNet50 [28]	83.67	87.34	68.71	77.39	71.22	77.66	85.65	86.93
Deeplab v3+ [37]	83.01	88.13	69.79	78.26	71.49	78.14	86.71	87.28
TransUNet [38]	84.95	90.07	71.92	80.23	74.89	80.41	88.21	89.36
DCSwin [39]	86.27	92.59	72.36	80.91	78.28	82.08	89.63	90.64
ABCNet [40]	85.63	91.27	72.38	81.04	77.71	81.61	89.51	90.33
CMTFNet [42]	86.31	92.06	72.81	81.99	80.23	82.68	90.43	90.78
UNetFormer [41]	86.61	**93.43**	72.52	81.25	78.94	82.55	90.02	90.59
VMamba [9]	88.01	92.78	73.64	81.65	80.56	83.33	91.89	91.31
RSVMamba (ours)	**88.69**	93.29	**74.51**	**82.47**	**81.72**	**84.14**	**92.76**	**92.13**

TABLE VII
EXPERIMENTAL RESULTS ON THE ISPRS POTSDAM DATASET, WITH THE BEST RESULTS INDICATED IN BOLD

Method	Imp. surface	Building	Low veg.	Tree	Car	mIoU(%)	AF(%)	OA(%)
FCN [35]	80.68	86.19	71.93	72.56	80.24	78.32	86.18	85.67
UNet [36]	81.82	88.23	72.41	73.85	81.94	79.65	86.72	86.53
ResNet50 [28]	82.30	89.12	74.95	76.76	82.38	81.10	89.24	87.97
Deeplab v3+ [37]	83.56	89.37	73.23	75.19	83.82	81.03	88.53	87.39
TransUNet [38]	85.21	92.19	75.50	79.32	85.63	83.57	90.06	88.94
DCSwin [39]	86.71	94.02	77.36	80.09	92.45	86.13	91.68	91.03
ABCNet [40]	86.79	93.14	76.75	79.31	92.06	85.61	91.41	90.87
CMTFNet [42]	87.13	94.44	77.51	80.16	92.32	86.31	92.33	91.21
UNetFormer [41]	87.51	94.29	77.18	80.07	92.56	86.36	92.05	91.62
VMamba [9]	88.02	94.58	78.51	80.32	92.95	86.88	92.38	91.24
RSVMamba (ours)	**88.71**	**95.19**	**79.29**	**80.81**	**93.47**	**87.49**	**92.96**	**91.87**

irregular edges, and blurred boundaries pose challenges for accurate classification. However, our model shows significant improvement for these two categories, which is one of the reasons we consider it suitable for TS classification.

To focus on these two categories relevant to this study, we conducted a set of experiments on the three-categories ISPRS Vaihingen dataset. The prediction results are shown in Table VIII, and the OA and category accuracy of all models show consistent improvement. For the standard evaluation metrics, mIoU, AF, and OA, the proposed model achieved results of 85.12%, 93.51%, and 95.17% on the three-categories ISPRS Vaihingen dataset. We believe this is mainly because the reduced number of categories led to a decrease in classification difficulty. All nonvegetation areas were consolidated into the category "Background," which reduced the impact of noise, allowing the model to focus on vegetation-related features during feature learning. The proposed RSVMamba achieved superior classification results in the category "Tree" compared to other models, with an IoU of 83.29%. This result demonstrates the potential of RSVMamba for application in TS classification tasks.

We visualized the prediction results on the ISPRS Vaihingen and ISPRS Potsdam datasets as illustrated in Figs. 7 and 8. It can be observed that most models exhibit good feature extraction capabilities for categories with large interclass differences and distinct features, such as "Building" and "Car," resulting in good overall segmentation results. Most models correctly classify pixels that were incorrectly labeled in the original labels as shown in the black dashed boxes in Fig. 7. However, for some special scenarios or specific categories, accurate recognition can be challenging. For instance, as shown in the red dashed boxes in Fig. 8, some pixels of the category "Tree" are sometimes incorrectly classified as "Low vegetation," and the incomplete cars at the edge of the image are often ignored and not accurately identified.

F. Module Analysis

1) Ablation Study: To evaluate the effects of the RK2 method and AutoDownsampling module on the classification results, we conducted an ablation study on the TS dataset. The third-to-last row of Table V presents the result of the model that uses the original zeroth-order hold rule and introduces

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TABLE VIII
EXPERIMENTAL RESULTS ON THE THREE-CATEGORIES ISPRS VAIHINGEN DATASET, WITH THE BEST RESULTS INDICATED IN BOLD

Method	Ba.d	Low veg.	Tree	mIoU(%)	AF(%)	OA(%)
FCN [35]	89.91	68.71	74.37	77.66	84.09	86.56
UNet [36]	91.35	69.62	77.03	79.33	84.97	87.93
ResNet50 [28]	92.77	70.01	78.28	80.35	87.13	89.24
Deeplab v3+ [37]	92.69	71.20	79.12	81.00	88.16	91.04
TransUNet [38]	94.24	72.83	81.19	82.75	89.33	92.15
DCSwin [39]	**96.99**	74.38	82.55	84.64	92.14	94.85
ABCNet [40]	95.77	73.85	81.96	83.86	90.62	93.81
CMTFNet [42]	96.18	74.43	82.32	84.31	91.40	94.01
UNetFormer [41]	96.35	73.95	82.18	84.16	91.35	94.29
VMamba [9]	96.51	74.63	82.61	84.58	92.62	94.22
RSVMamba (ours)	96.91	**75.15**	**83.29**	**85.12**	**93.51**	**95.17**

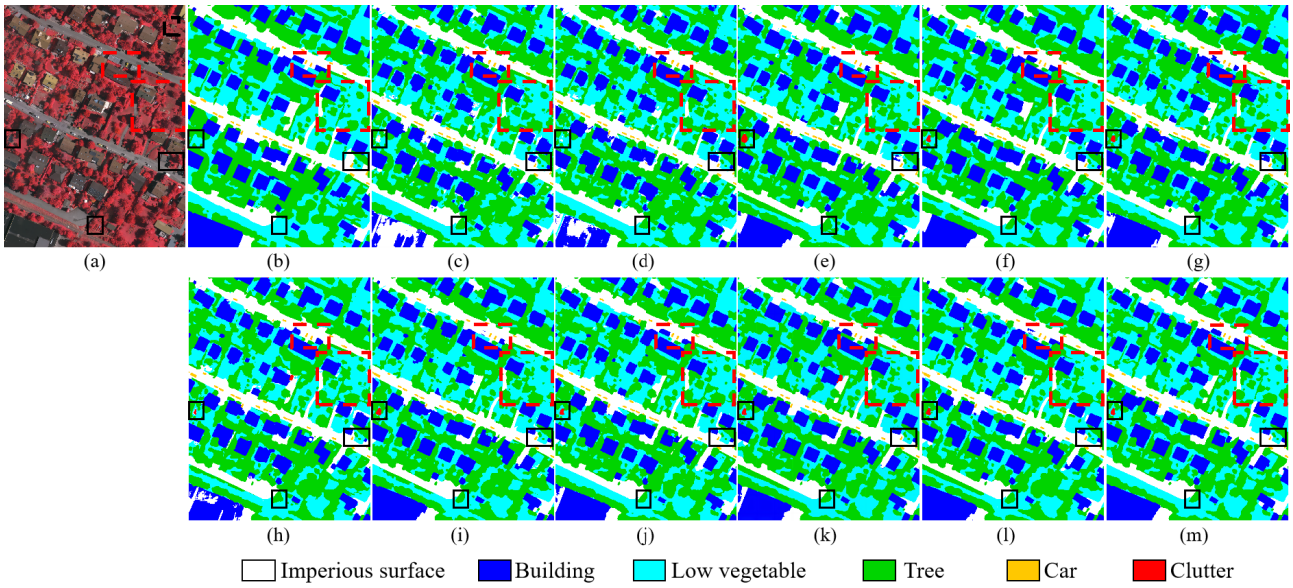


Fig. 7. Visualization results of the ISPRS Vaihingen dataset. (a) Original image. (b) Ground truth. (c) FCN [35]. (d) UNet [36]. (e) ResNet50 [28]. (f) Deeplab v3+ [37]. (g) TransUNet [38]. (h) DCSwin [39]. (i) ABCNet [40]. (j) Unetformer [41]. (k) CMTFNet [42]. (l) VMamba [9]. (m) RSVMamba.

the AutoDownsampling module to VMamba. The second-to-last row shows the result of the model that uses the RK2 method to discretize the state-space model of VMamba. It can be observed that both AutoDownsampling and RK2 have an explicit role to play in improving model performance. We believe that the RK2 method accumulates less error and maintains higher accuracy by evaluating the system derivatives twice. It approximates the continuous dynamic behavior of the system more accurately than the zeroth-order hold rule, providing better responses to changes in the system state and capturing dynamic characteristics more effectively, thus improving the overall classification accuracy. In the case of AutoDownsampling, although the insertion of this module did not significantly improve the OA, it had a positive impact. The classification accuracy of some sparsely distributed categories "Ph.e" and "Pa.a" improved significantly, with their mIoU increasing notably. We believe this method effectively focuses on important information that occupies relatively small spatial positions. The AutoDownsampling module dynamically evaluates the importance of each token and adaptively adjusts the sampling stride. It captures the intrinsic importance of features and preserves spatial structural information, enabling the model to better adapt to local variations in the data and avoid the loss of crucial information. Therefore, the classification accuracy of TS that are scattered within large pure forests has been significantly improved.

2) Computational Cost Evaluation: We evaluated the computational overhead of the AutoDownsampling module and analyzed its inference time and floating-point operations (FLOPs) under different input image sizes and values of K (the number of nearest neighbors), allowing for reasonable adjustments based on computational resources and accuracy requirements. As shown in Table IX, the inference time and FLOPs tend to increase as the input image size

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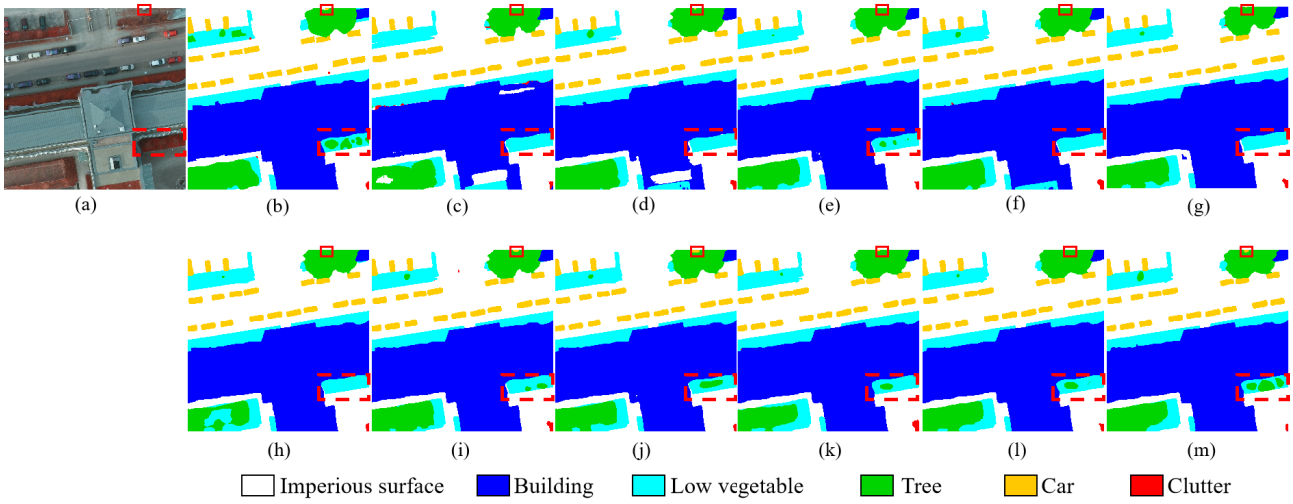


Fig. 8. Visualization results of the ISPRS Potsdam dataset. (a) Original image. (b) Ground truth. (c) FCN [35]. (d) UNet [36]. (e) ResNet50 [28]. (f) Deeplab v3+ [37]. (g) TransUNet [38]. (h) DCSwin [39]. (i) ABCNet [40]. (j) Unetformer [41]. (k) CMTFNet [42]. (l) VMamba [9]. (m) RSVMamba.

TABLE IX
COMPUTATIONAL COMPLEXITY TESTING OF AUTODOWNSAMPLING MODULES

Size	K	Time(ms)	FLOPs(M)	Pi.s	Cu.l	Qu.s	Li.f	Eu.g	Ph.e	Ca.s	Pa.a	Ot.b	De.t	Ba.d	mIoU(%)
512^2	**1**	19.67	4.92	66.71	75.30	51.21	73.66	85.13	80.89	80.33	66.81	60.91	44.43	86.31	70.15
	3	21.27	21.07	66.87	75.51	51.35	73.93	85.59	81.46	80.31	66.97	**60.98**	44.62	86.39	70.36
	5	22.57	44.25	66.89	75.62	**51.41**	74.17	85.85	81.75	80.37	67.16	60.95	44.85	**86.46**	70.50
	7	23.69	69.93	**66.95**	**75.73**	51.39	**74.36**	**85.98**	**81.99**	**80.44**	**67.38**	60.89	**44.96**	86.43	**70.59**
1024^2	**1**	66.23	20.46	67.46	75.91	51.82	74.34	86.01	81.79	80.61	67.47	61.52	46.04	86.53	70.86
	3	73.29	84.41	67.69	76.08	51.97	74.76	86.28	82.32	80.74	67.75	61.67	46.27	86.81	71.12
	5	77.75	154.71	67.75	76.16	**52.09**	75.11	86.37	82.55	**80.81**	67.83	**61.71**	**46.31**	86.78	71.22
	7	81.76	239.71	**67.86**	**76.27**	52.07	**75.23**	**86.45**	**82.81**	80.76	**67.98**	61.66	46.29	**86.87**	**71.30**

increases. While a larger input size aids in capturing finer details, especially for complex tasks such as TS classification, it also increases memory requirements and computational load. Therefore, users may select appropriate input sizes according to the available computational resources and accuracy requirements in practical applications. Similarly, as the value of K increases, the inference time and FLOPs also increase. A larger K value represents a larger neighborhood, enabling the capture of more contextual information, especially in edges, details, and complex textures. Therefore, it will greatly improve the classification accuracy of categories such as "Ph.e" and "Pa.a," which are sparsely distributed and dispersed within large areas of pure forest, reducing the possibility of these categories being overlooked, as they are easily submerged in other categories. However, the accuracy improvement tends to plateau as K continues to increase. Accordingly, we selected $K = 3$ in our application scenario to achieve a balance between accuracy and computational efficiency. Given the computational complexity of the AutoDownsampling module, it is applied only to the last layer of the RSVMamba, where the feature map has been downsampled to 32×32, this can maximize the performance benefits of the module while controlling computational overhead.

3) Stability Analysis: To evaluate the stability of the RK2 method in different scenarios, we constructed the following three datasets reflecting diverse conditions and data distributions that may be encountered in real-world applications.

1) *Low-Complexity Data (TS1):* This dataset comprises 70 images selected from the original TS dataset under simple environmental conditions, with no shadows or occlusions, and uniform TS distributions. It includes six categories "Cu.l," "Li.f," "Eu.g," "Ca.s," "Ot.b," and "Ba.d," aiming to verify the stability of the RK2 method in relatively simple scenarios.
2) *High-Complexity Data (TS2):* This dataset consists of 100 images selected from the original TS dataset under complex environmental conditions, including terrain variations, shadows, and mixed TS distributions. It contains the same 11 categories as the original TS dataset and is designed to verify the stability of the RK2 method in complex scenarios.
3) *Imbalanced Distribution Data (TS3):* This dataset comprises 60 images selected from the original TS dataset with imbalanced distributions, primarily large areas of pure forest with scattered other TS occupying sparse spatial locations. It includes five categories "Pi.s," "Cu.l,"

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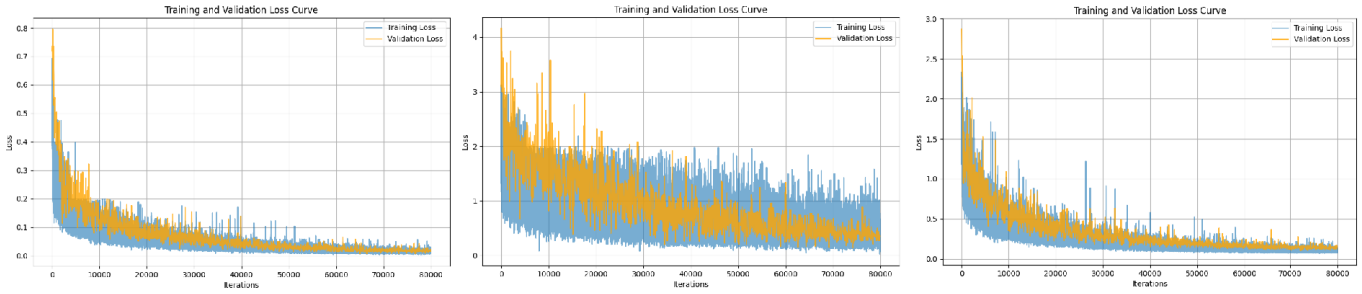


Fig. 9. Training and validation loss curves. (a) Loss curve of the TS1 dataset. (b) Loss curve of the TS2 dataset. (c) Loss curve of the TS3 dataset.

"Li.f," "Ph.e," and "Ba.d." Since it is difficult to ensure that only these four species are included in each image, the other TS that appear in the image are grouped into the category "Ba.d," to assess whether the RK2 method maintains stability in scenarios with imbalanced data distributions.

As shown in Fig. 9, across datasets of varying complexities and data distributions, although there is a certain degree of oscillation, it gradually diminishes without exhibiting characteristics of numerical divergence. Both the training losses and validation losses exhibit a consistent convergence trend overall. This consistency in convergence across different scenarios demonstrates the stability of the RK2 method.

G. Limitations and Future Work

While RSVMamba demonstrated effective performance across multiple datasets and achieved acceptable accuracy in classifying multiple TS within complex heterogeneous forests, there are still some limitations in real-world deployments. First, the complexity of TS classification leads to the requirement of a substantial amount of annotated data. We manually annotated the TS dataset through visual interpretation, cross-checked by at least one additional interpreter, and the final annotation results were reviewed and corrected by an expert to ensure accuracy. This process is both time-consuming and labor-intensive, which is the major limitation for broader applications. Thus, in the future, we will explore methods that reduce dependency on annotated data, such as semi-supervised or unsupervised learning methods. Second, with an input size of 1024×1024, the inference time of the proposed model is 279.41 ms. The classification of high-spatial-resolution images generally entails significant computational requirements, posing challenges for large-scale data processing. In this regard, future work will focus on model optimization and exploring lightweight versions to reduce computational demands without compromising accuracy. Additionally, RSVMamba was primarily developed to classify multiple TS within subtropical forests located in the study area. Due to dataset limitations, the model has not been validated on other forest types or species. While benchmark datasets show promising results, unforeseen challenges may arise when applying the model to real-world applications of different scales and diverse forest environments. In future work, we aim to assess the scalability and robustness of the model in diverse, real-world environments to ensure its broad applicability.

VI. Conclusion

In this work, we collected UAV RGB images of a subtropical forest located in Qimen County, Huangshan City, Anhui Province, China, and constructed a TS classification dataset. Moreover, we proposed a promising TS classification model, which is an effective framework for remote sensing image semantic segmentation inspired by the recent VMamba architecture. To improve and tailor VMamba for the task at hand, a higher-order discretization method was introduced to discretize the state-space model, improving the model's accuracy and stability. Additionally, considering that forest TS are often unevenly distributed, with sporadically distributed species often being overlooked and inaccurately identified, we employed an adaptive downsampling method to focus on the species that are submerged within a large number of other species. We validated the generalization capability of our proposed model on the publicly available ISPRS Vaihingen and Potsdam datasets, demonstrating its effectiveness in handling data from diverse environments and distributions.

Our method relies solely on single-temporal RGB images from a practical forestry application perspective, addressing the data acquisition problem in the current TS classification task to a certain extent. Compared to the high acquisition costs and complex preprocessing of hyperspectral data, and the real-time limitations of multitemporal RGB data, this method offers a more cost-effective and flexible approach for forest TS classification, providing a practical solution for obtaining information on TS composition and spatial distribution. This study demonstrates the critical role of UAV RGB images in forest TS classification, and the potential for synergistic use of high-resolution UAV images and deep learning, providing promising prospects for large-scale forestry research.

Acknowledgment

The authors would like to thank the International Society for Photogrammetry and Remote Sensing for acquiring and providing the public datasets used in this study.

References

[1] A. Harikumar, C. Paris, F. Bovolo, and L. Bruzzone, "A crown quantization-based approach to tree-species classification using high-density airborne laser scanning data," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 4444–4453, May 2021.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:57:14 UTC from IEEE Xplore. Restrictions apply.

[2] F. E. Fassnacht et al., "Review of studies on tree species classification from remotely sensed data," *Remote Sens. Environ.*, vol. 186, pp. 64–87, Dec. 2016.

[3] Y. Li, H. Zhang, X. Xue, Y. Jiang, and Q. Shen, "Deep learning for remote sensing image classification: A survey," *Wiley Interdiscipl. Rev., Data Mining Knowl. Discovery*, vol. 8, no. 6, p. 1264, May 2018.

[4] S. Li, W. Song, L. Fang, Y. Chen, P. Ghamisi, and J. A. Benediktsson, "Deep learning for hyperspectral image classification: An overview," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6690–6709, Sep. 2019.

[5] J. A. Pereira et al., "Unmanned aerial vehicle surveys reveal unexpectedly high density of a threatened deer in a plantation forestry landscape," *Oryx*, vol. 57, no. 1, pp. 89–97, Jan. 2023.

[6] M. Gazzea, L. M. Kristensen, F. Pirotti, E. E. Ozguven, and R. Arghandeh, "Tree species classification using high-resolution satellite imagery and weakly supervised learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4414311.

[7] J. Komárek, "The perspective of unmanned aerial systems in forest management: Do we really need such details?" *Appl. Vegetation Sci.*, vol. 23, no. 4, pp. 718–721, Oct. 2020.

[8] M. P. Ferreira, F. H. Wagner, L. E. O. C. Arag ao, Y. E. Shimabukuro, and C. R. de Souza Filho, "Tree species classification in tropical forests using visible to shortwave infrared WorldView-3 images and texture analysis," *ISPRS J. Photogramm. Remote Sens.*, vol. 149, pp. 119–131, Mar. 2019.

[9] Y. Liu et al., "VMamba: Visual state space model," 2024, *arXiv:2401.10166*.

[10] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[11] A. Rößler, "Second order Runge–Kutta methods for Itô stochastic differential equations," *SIAM J. Numer. Anal.*, vol. 47, no. 3, pp. 1713–1738, 2009.

[12] ISPRS. (2024). *2D Semantic Labeling—Vaihingen Data*. [Online]. Available: https://www.isprs.org/education/benchmarks/UrbanSemLab/2d-sem-label-vaihingen.aspx

[13] ISPRS. (2024). *2D Semantic Labeling Contest—Potsdam*. [Online]. Available: https://www.isprs.org/education/benchmarks/UrbanSemLab/2d-sem-label-potsdam.aspx

[14] R. Pu, "Mapping tree species using advanced remote sensing technologies: A state-of-the-art review and perspective," *J. Remote Sens.*, Nov. 2021, Art. no. 9812624, doi: 10.34133/2021/9812624.

[15] L. E. C. La Rosa, C. Sothe, R. Q. Feitosa, C. M. de Almeida, M. B. Schimalski, and D. A. B. Oliveira, "Multi-task fully convolutional network for tree species mapping in dense forests using small training hyperspectral data," *ISPRS J. Photogramm. Remote Sens.*, vol. 179, pp. 35–49, Sep. 2021.

[16] L. Chen, J. Wu, Y. Xie, E. Chen, and X. Zhang, "Discriminative feature constraints via supervised contrastive learning for few-shot forest tree species classification using airborne hyperspectral images," *Remote Sens. Environ.*, vol. 295, Sep. 2023, Art. no. 113710.

[17] L. Chen, X. Tian, G. Chai, X. Zhang, and E. Chen, "A new CBAM-P-Net model for few-shot forest species classification using airborne hyperspectral images," *Remote Sens.*, vol. 13, no. 7, p. 1269, Mar. 2021.

[18] J. Mäyrä et al., "Tree species classification from airborne hyperspectral and LiDAR data using 3D convolutional neural networks," *Remote Sens. Environ.*, vol. 256, Apr. 2021, Art. no. 112322.

[19] S. Nezami, E. Khoramshahi, O. Nevalainen, I. Pölönen, and E. Honkavaara, "Tree species classification of drone hyperspectral and RGB imagery with deep learning convolutional neural networks," *Remote Sens.*, vol. 12, no. 7, p. 1070, Mar. 2020.

[20] I. Pölönen et al., "Tree species identification using 3D spectral data and 3D convolutional neural network," in *Proc. 9th Workshop Hyperspectral Image Signal Process., Evol. Remote Sens. (WHISPERS)*, Sep. 2018, pp. 1–5.

[21] B. Zhang, L. Zhao, and X. Zhang, "Three-dimensional convolutional neural network model for tree species classification using airborne hyperspectral images," *Remote Sens. Environ.*, vol. 247, Sep. 2020, Art. no. 111938.

[22] H. Qin, W. Zhou, Y. Yao, and W. Wang, "Individual tree segmentation and tree species classification in subtropical broadleaf forests using UAV-based LiDAR, hyperspectral, and ultrahigh-resolution RGB data," *Remote Sens. Environ.*, vol. 280, Oct. 2022, Art. no. 113143.

[23] S. Natesan, C. Armenakis, and U. Vepakomma, "ResNet-based tree species classification using UAV images," *Int. Arch. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 42, pp. 475–481, Jun. 2019.

[24] S. Egli and M. Höpke, "CNN-based tree species classification using high resolution RGB image data from automated UAV observations," *Remote Sens.*, vol. 12, no. 23, p. 3892, Nov. 2020.

[25] Y. Li, G. Chai, Y. Wang, L. Lei, and X. Zhang, "ACE R-CNN: An attention complementary and edge detection-based instance segmentation algorithm for individual tree species identification using UAV RGB images and LiDAR data," *Remote Sens.*, vol. 14, no. 13, p. 3035, Jun. 2022.

[26] R. Pierdicca, L. Nepi, A. Mancini, E. S. Malinverni, and M. Balestra, "UAV4Tree: Deep learning-based system for automatic classification of tree species using RGB optical images obtained by an unmanned aerial vehicle," *ISPRS Ann. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 10, pp. 1089–1096, Dec. 2023.

[27] S. Natesan, C. Armenakis, and U. Vepakomma, "Individual tree species identification using dense convolutional network (DenseNet) on multitemporal RGB images from UAV," *J. Unmanned Vehicle Syst.*, vol. 8, no. 4, pp. 310–333, Dec. 2020.

[28] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[29] M. L. Clark, D. A. Roberts, and D. B. Clark, "Hyperspectral discrimination of tropical rain forest tree species at leaf to crown scales," *Remote Sens. Environ.*, vol. 96, nos. 3–4, pp. 375–398, Jun. 2005.

[30] S. Hartling, V. Sagan, and M. Maimaitijiang, "Urban tree species classification using UAV-based multi-sensor data fusion and machine learning," *GIScience Remote Sens.*, vol. 58, no. 8, pp. 1250–1275, Nov. 2021.

[31] N. Li, S. Jiang, J. Xue, S. Ye, and S. Jia, "Texture-aware self-attention model for hyperspectral tree species classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5502215.

[32] L. Chen, Y. Wei, Z. Yao, E. Chen, and X. Zhang, "Data augmentation in prototypical networks for forest tree species classification using airborne hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4410116.

[33] C. Szegedy, V. Vanhoucke, S. Ioffe, J. Shlens, and Z. Wojna, "Rethinking the inception architecture for computer vision," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 2818–2826.

[34] G. Huang, Z. Liu, L. Van Der Maaten, and K. Q. Weinberger, "Densely connected convolutional networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2017, pp. 4700–4708.

[35] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3431–3440.

[36] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. 18th Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.* Cham, Switzerland: Springer, 2015, pp. 234–241.

[37] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam, "Encoder–decoder with Atrous separable convolution for semantic image segmentation," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 801–818.

[38] J. Chen et al., "TransUNet: Transformers make strong encoders for medical image segmentation," 2021, *arXiv:2102.04306*.

[39] L. Wang, R. Li, C. Duan, C. Zhang, X. Meng, and S. Fang, "A novel transformer based semantic segmentation scheme for fine-resolution remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[40] R. Li, S. Zheng, C. Zhang, C. Duan, L. Wang, and P. M. Atkinson, "ABCNet: Attentive bilateral contextual network for efficient semantic segmentation of fine-resolution remotely sensed imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 181, pp. 84–98, Nov. 2021.

[41] L. Wang et al., "UNetFormer: A UNet-like transformer for efficient semantic segmentation of remote sensing urban scene imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 190, pp. 196–214, Mar. 2022.

[42] H. Wu, P. Huang, M. Zhang, W. Tang, and X. Yu, "CMTFNet: CNN and multiscale transformer fusion network for remote-sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 2004612.

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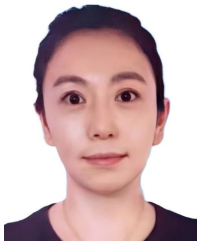
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